# Brenner - Final Project

## The Fish Who Cried Wolf: What the Atlantic Wolffish Tells Us About the Future of the Gulf of Maine

### Introduction

*Anarhichas lupus*, commonly known as the Atlantic Wolffish, is a large carnivorous demersal fish that spends most of its life feeding on benthic invertebrates such as crabs, scallops, urchins, and lobsters in the Northern Atlantic Ocean (Bianucci et al., 2016). Its position as a secondary consumer helps prevent invasive species, such as *Cancer maenas* (green crabs), and grazers, such as *Strongylocentrotus droebachiensis* (green sea urchins), from overconsuming seagrass, making the wolffish a fierce protector of the seagrass habitat. A generally sedentary species, the wolffish inhabits rocky burrows and crevices, making it hard to find in large bottom trawl surveys conducted in the spring and fall, leading to its classification as a data-deficient species (Fairchild et al., 2015).

Normally caught as bycatch, despite its minimal economic importance as a fishery in the Gulf of Maine, attempts have been made to cultivate wolffish populations for aquaculture. Found in temperatures ranging from -1º to 10ºc, the wolffish's ability to synthesize antifreeze proteins makes it a prime candidate for sea cage farming in colder climates, which are considered unsuitable for many other marine species (Le François et al., 2004). Further research has investigated the aquaculture potential of the Atlantic wolffish, examining its relatively moderate stress physiology by exposing it to handling stress in the aquaculture environment, making it a prime candidate for future cultivation (Hedén et al., 2025).

Although not necessarily a direct competitor with humans in common fisheries such as scallops and lobsters, there is significant overlap in the hunting grounds of both the Atlantic wolffish and humans. Often pulled up in lobster traps, the Atlantic wolffish will eat the trapped lobsters, leaving only lobster carcasses behind and a menacing bite for lobstermen to deal with. This overlap between wolffish and humans, in dredging zones for scallops and trapping areas for lobsters, can negatively affect wolffish habitat, further solidifying its place as a species of concern under the United States Endangered Species Act (Bianucci et al., 2016). Although it does not receive the same public attention as more commercially important marine species, the Atlantic wolffish remains an important study species. Its role as a protector of kelp forests and seagrass habitats provides insight into the health of the North Atlantic ecosystem. Furthermore, the aquaculture potential of the Atlantic wolffish underscores the importance of studying its habitat preferences in the search for more sustainable fishing practices.

This study examines the current and future population dynamics of the Atlantic wolffish in the Gulf of Maine. Using both nowcasts and forecasts of the species distribution within the Gulf of Maine, one can gain insight into the health of key kelp forest and seagrass habitats, highlighting areas for increased conservation interventions to prevent wolffish habitat destruction from trawling. These species distribution forecasts can also inform policy decisions for commercial lobster and scallop fisheries, preventing accidental bycatch of wolffish and protecting kelp ecosystems and species of greater commercial importance that dwell within them.

### Examining the Data

Data for creating nowcasts and forecasts of the Atlantic wolffish species distributions was extracted from the Ocean Biodiversity Information System (OBIS). This data consisted of observations from a variety of sources, including scientific research programs, citizen science initiatives (such as iNaturalist), and individual observations in which a wolffish was spotted and reported to a larger database. Datapoints represent where an individual spotted the wolffish, but the lack of data does not necessarily indicate the absence of the wolffish. This data set totaled 7009 unfiltered observations in the Gulf of Maine; however, many data points lacked the number of individuals counted per observation or the date they were spotted. To control for this lack of data, the data were filtered to complete observations from after 1970 (an arbitrary cutoff), leaving a total of 1162 observations of wolffish in the Gulf of Maine.

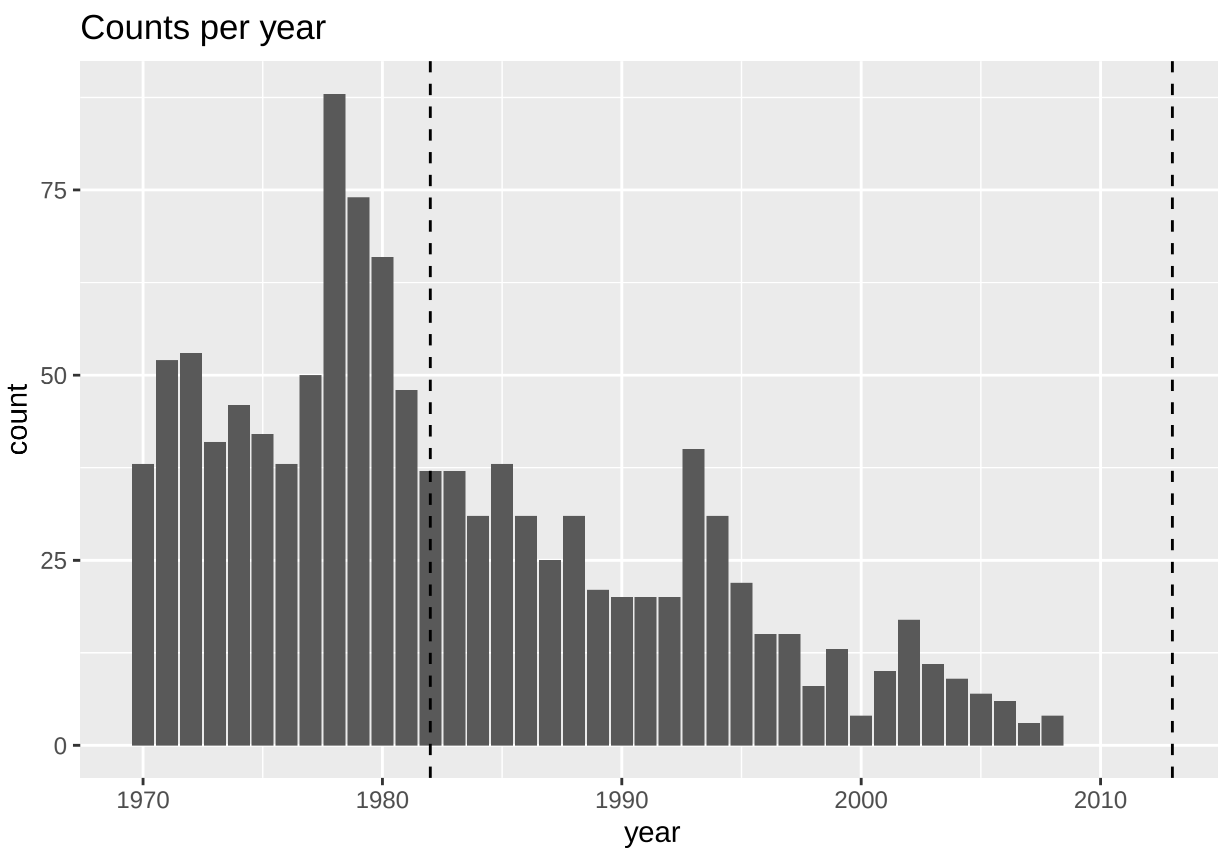


Figure 1: Figure 1) Histogram of annual filtered Atlantic wolffish observation data in the Gulf of Maine from 1970 to 2009 (n = 1162).

Examining observation counts by year, there is a significant decrease in the number of Atlantic wolffish observed from 1970 to 2009, with a maximum of 88 in 1978. However, following 2009, there were zero recorded observations that included the individual count. This could be attributed to a possible change in the type of observation contributed, where individuals using individual observations (which constituted the majority of the observations) did not report the specific number of wolffish spotted. This lack of contemporary data excludes distribution information from the years since the 2013 heatwave and the continuously warming climate since.

Examining the observation counts by month, the Atlantic wolffish's status as a data-deficient species further comes into view, as there are no observations for the months of January and June. In fact, much of the data is compiled in the spring and fall months, with a sharp increase in observations in April, when the National Oceanic and Atmospheric Administration (NOAA) conducts bottom trawl surveys. Since the wolffish is a bottom-dwelling fish, it is hard for the general public to spot from the surface and is also difficult to reach with the trawls used in surveys due to its preference for rocky habitats (Fairchild et al., 2015).

Their spatial distribution, as shown on the bias map, demonstrates clustering of wolffish along the coast and on banks throughout the Gulf of Maine. In these areas, increased nutrient upwelling from colder currents supports a thriving ecosystem for a variety of marine species, especially scallops. At Stellwagen Bank, the

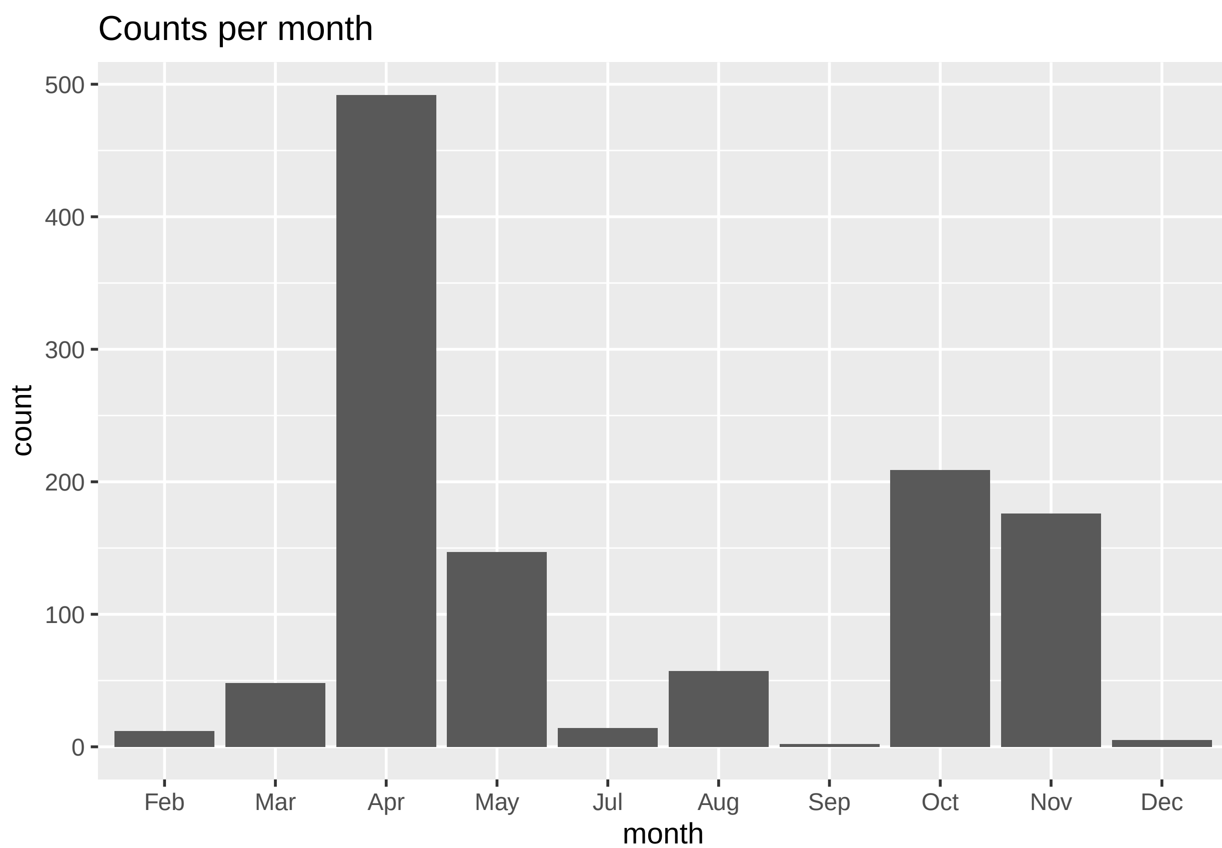


Figure 2: Figure 2) Histogram of monthly filtered Atlantic wolffish observation data in the Gulf of Maine from January to December (n = 1162).

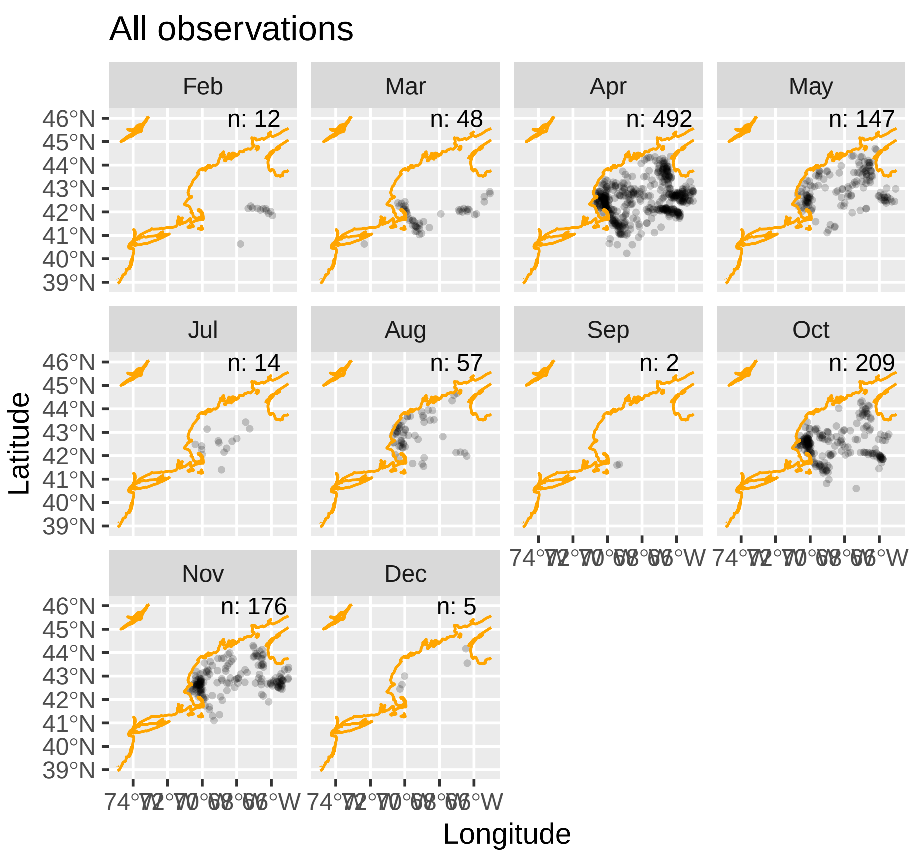


Figure 3: Figure 3) Map of monthly Atlantic wolffish observation distribution in the Gulf of Maine (n = 1162).

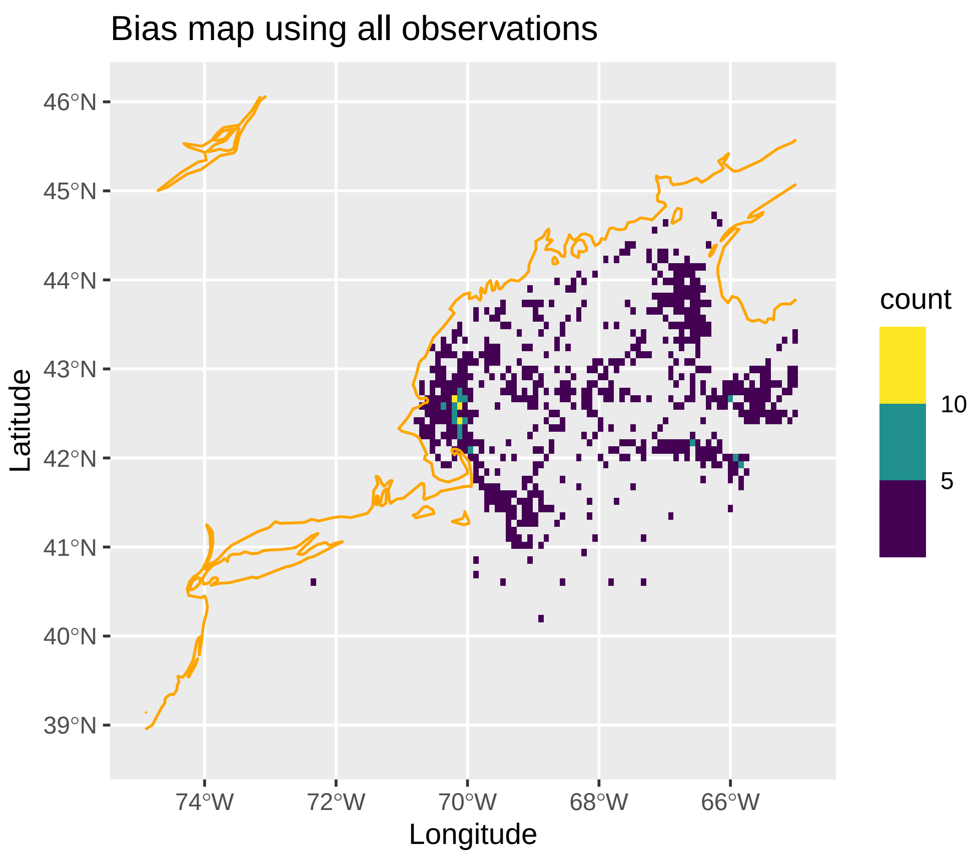


Figure 4: Figure 4) Spatial bias map of Atlantic wolffish observation distribution in the Gulf of Maine (n = 1162).

increase in nestled scallops on the seafloor has been found to attract foraging Atlantic wolffish in the area (Fairchild et al., 2015). This phenomenon could extend to other banks, such as George's Bank southwest of Cape Cod, Brown's Bank off the southwest coast of Nova Scotia, and other coastal areas.

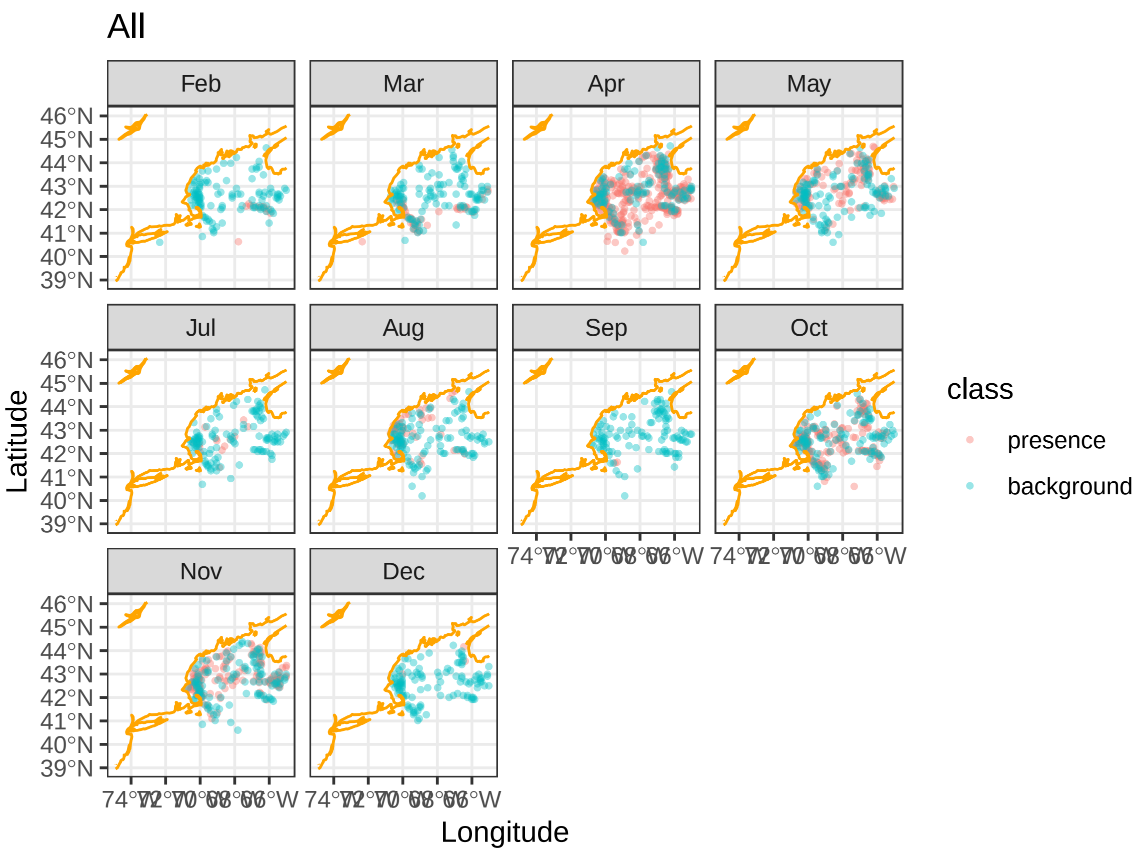


Figure 5: Figure 5) Map of monthly presence (observation) and background (pseudo-absence) data of Atlantic wolffish in the Gulf of Maine (n = )

After examining the data, background points were randomly sampled to balance the number of observations and match the regional preferences of the wolffish presence data. These background points serve as pseudo-absences in the dataset, as it is impossible to determine the exact distribution of the Atlantic wolffish. The environmental covariate data used for each point, both presence and background, come from the Brickman dataset, which uses data from the Bedford Institute of Oceanography North Atlantic Model and the Regional Ocean Modeling System to forecast the ocean conditions of the Gulf of Maine (Brickman et al., 2021). These covariates include depth, mixed layer depth, bottom salinity, surface salinity, surface temperature, bottom temperature, and current vectors. Generally, the background and presence points inhabit areas with similar ocean conditions, with the presence data showing a skewed distribution toward colder sea surface temperatures and shallower mixed layer depths.

## Modeling

After all the data were collected and cleaned, the data were spatially thinned to account for spatial bias along the banks. The data was then split into training data (1632 data points) to build the models and testing data (471 data points) to evaluate the models' accuracy. Workflows for generalized linear, random forest, boosted regression tree, and maximum entropy models were then defined and deployed using machine learning to determine the correct hyperparameters. The success of each model iteration was determined using various metrics, including the continuous Boyce index, area under the receiver operating characteristic (ROC) curve,

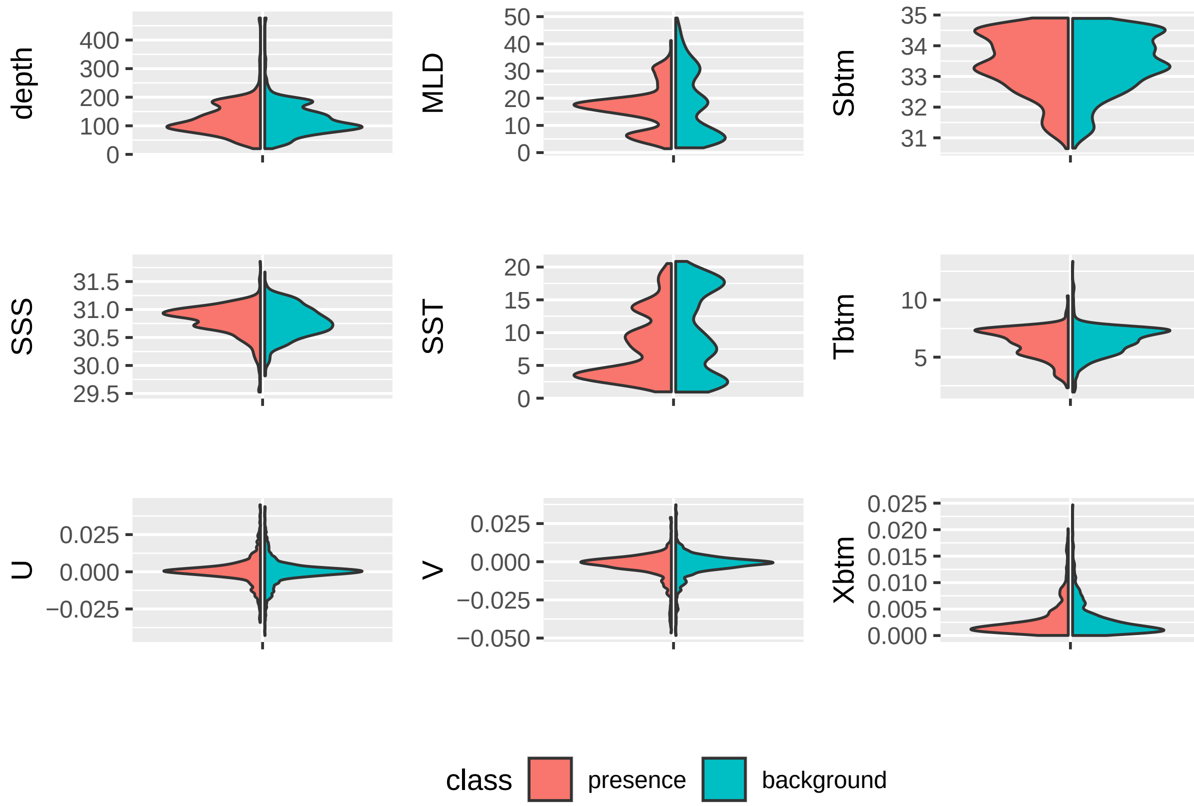


Figure 6: Figure 6) Violin plots comparing presence and background distribution across various covariates., fig.align='center'


Figure 7: Figure 7) Map of Atlantic wolffish observation distribution split between training and testing datasets.

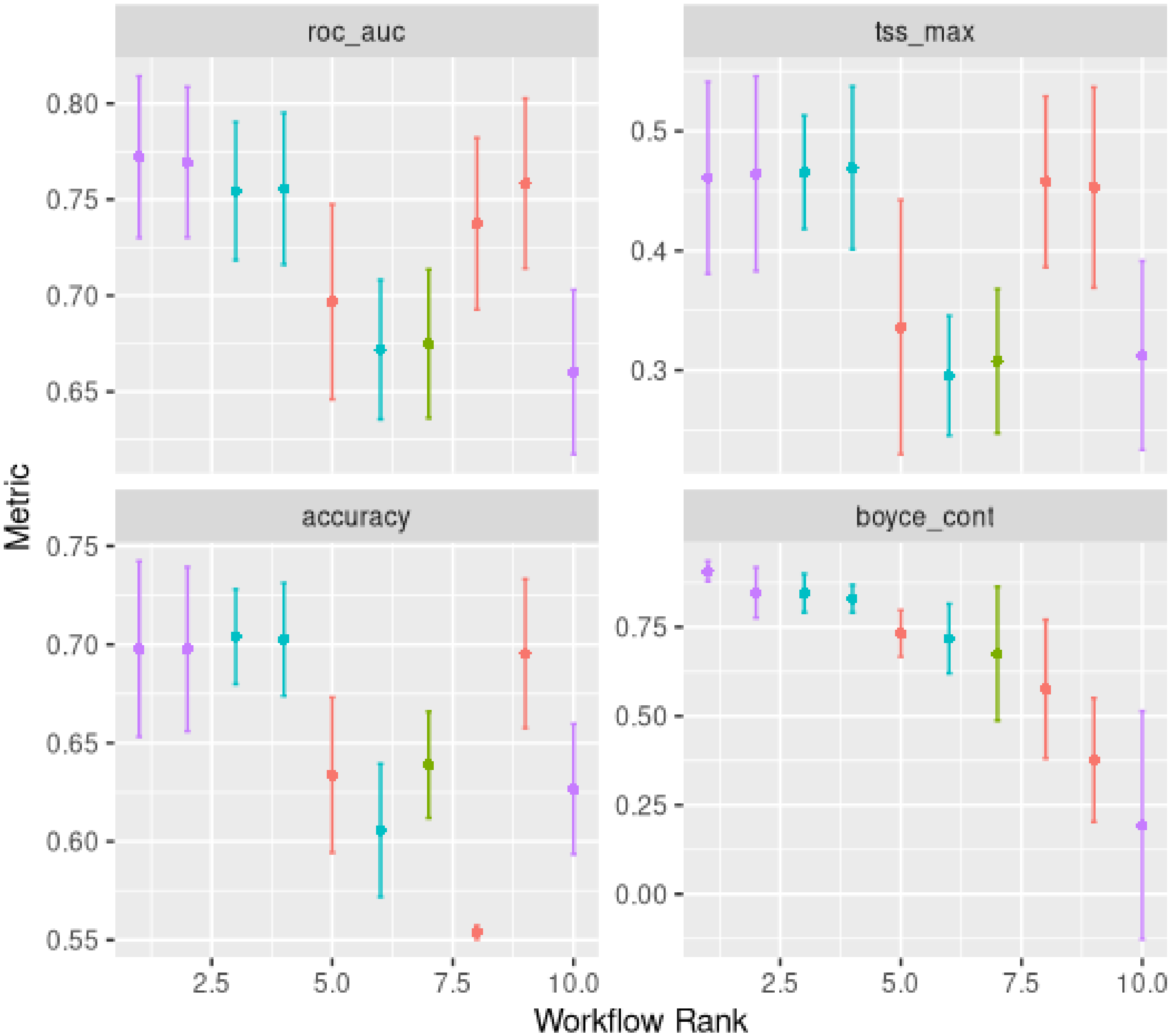


Figure 8: Figure 8) Plots of model success metrics for four different distribution models (boosted tree, generalized linear, maximum entropy, and random forest models) and their variants.

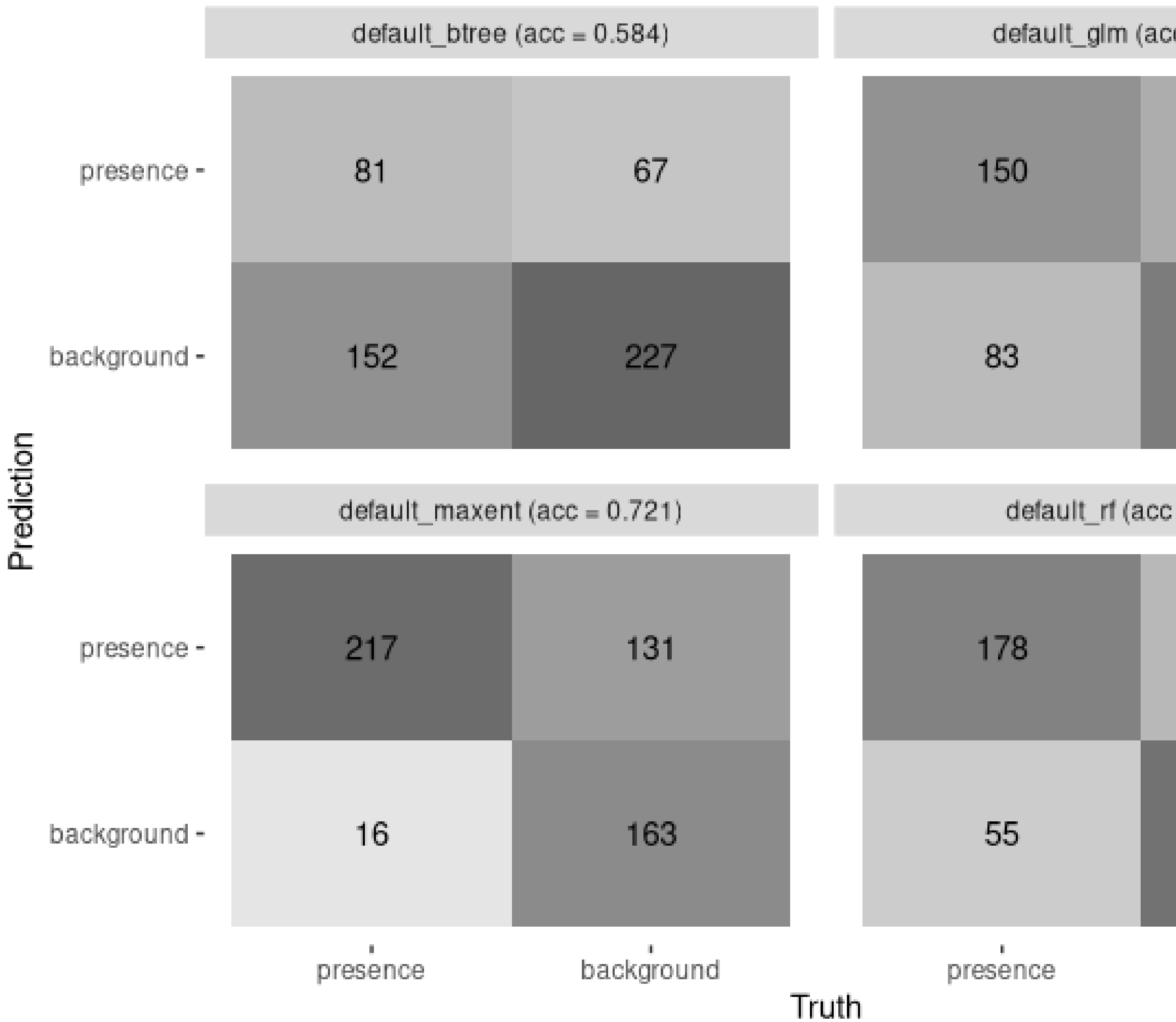


Figure 9: Figure 9) Confusion matrix of four different distribution models (boosted tree, generalized linear, maximum entropy, and random forest models) comparing model prediction of presence vs. background data against the truth.

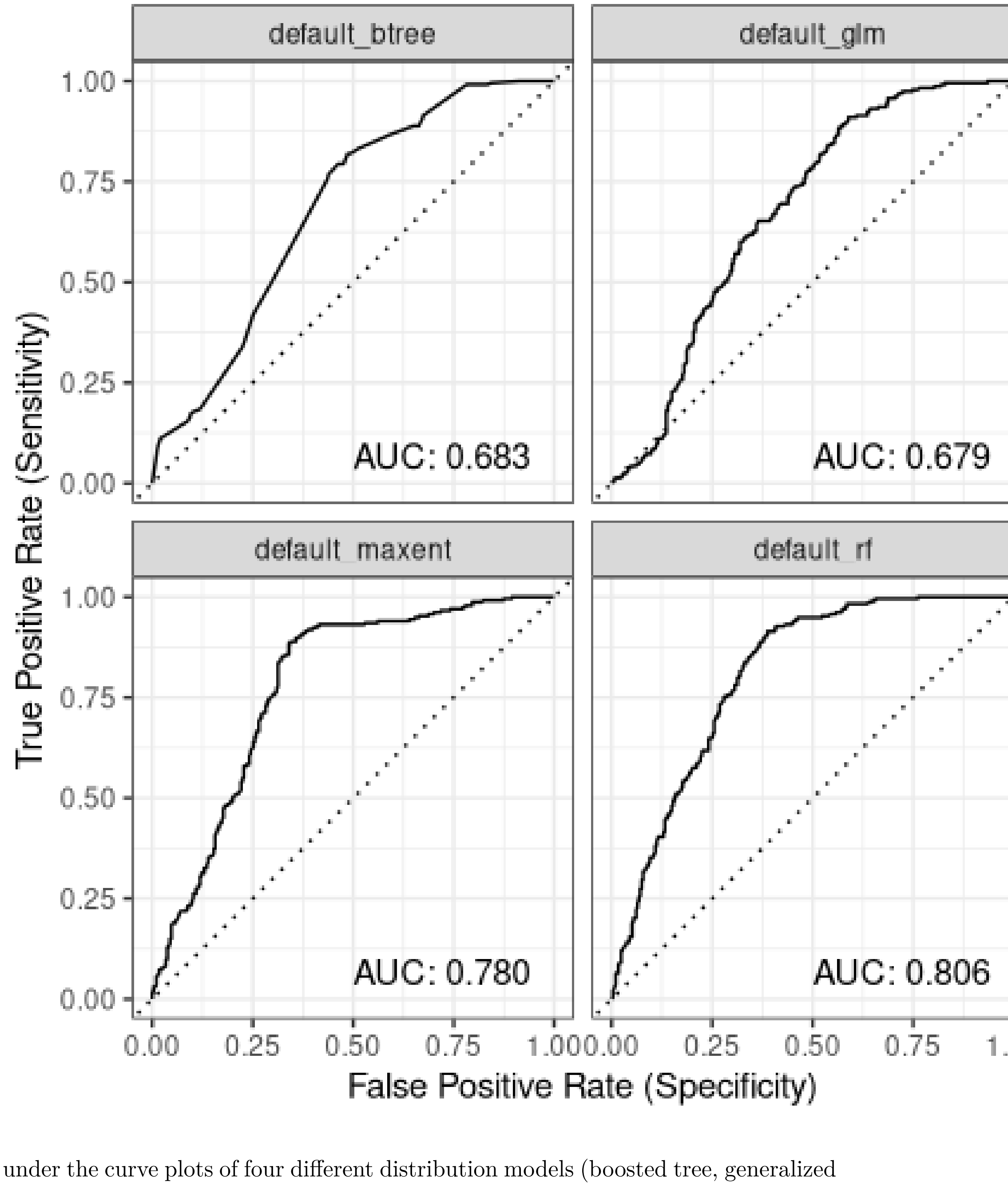


Figure 10: Figure 10) Area under the curve plots of four different distribution models (boosted tree, generalized linear, maximum entropy, and random forest models) comparing model certainty.

training stress score, and accuracy metrics. After the best model was selected, each model was compared to the others using the same metrics and a confusion matrix. These metrics showed that the random forest model performed best at modeling the distribution of the Atlantic wolffish population in the Gulf of Maine.

Examining the importance of each covariate in the random forest model's outputs, it appears that sea surface temperature and mixed layer depth are significant in determining the distribution of wolffish. Although the Atlantic wolffish is typically bottom-dwelling, the sea surface temperature's importance contributes significantly more than bottom temperature to the random forest model. This could be due to the higher resolution and accuracy of sea surface temperature in the Brickman dataset, with sea surface temperature still correlating with bottom temperature. This increased contribution of temperature to model output is corroborated by a study by Hinchcliffe et al. (2025), which examined acute and chronic exposure to warmer water temperatures and found decreased intestinal barrier function, reduced growth, and altered metabolic rates. Mixed layer depth also influences ocean-floor temperature, so Atlantic wolffish might prefer a colder habitat with a higher mixed layer depth. Conversely, salinity's lack of importance in the model could result from the haloplasticity of the Atlantic wolffish, which exhibits extreme tolerance to higher salinities (Le François et al., 2004).

## Forecasts

After deciding to employ the random forest model, one nowcast and four forecasts were generated to examine the distribution of the Atlantic Wolffish in 2025, 2055, and 2075. The forecasts exhibited outcomes for two different scenarios: RCP 4.5 (where greenhouse gas emissions are mitigated) and RCP 8.5 (where greenhouse gas emissions remain as is). These projections of Atlantic wolffish distribution display an overall northward shift in the population, with the wolffish retreating into the Bay of Fundy. There is also a seasonal shift in which the presence of wolffish increases in colder months, with the vernal peak shifting from April to March; conversely, the autumnal peak appears to shift towards warmer months, from October to September. These distributional and seasonal shifts are further exacerbated in forecasts that incorporate RCP 8.5 assumptions, in which the Gulf of Maine experiences more intense warming.

Despite this suggested shift, the forecasting results for data-poor months conflict with those for data-rich months, and the model's contribution to the model, where Atlantic wolffish tend to prefer colder water temperatures. Furthermore, the forecasts seem to place the wolffish presence consistently off-shore far away from George's bank, also conflicting with their known habitat preferences of rocky banks. Since the model does not take sea floor substrate into account, this phenomenon could be the result of similarities in mixed layer depth of the Gulf Stream off of George's bank. Also, the sedentary nature of the Atlantic wolffish is not reflected in this model, showing large migration patterns in both forecasted and nowcasted projections. Although the models' accuracy is questionable, there is still much to be learned from these forecasts of Atlantic wolffish presence in the Gulf of Maine.

## Discussion

Regardless of discrepancies in the random forest model of Atlantic wolffish population distribution, the forecasts' northerly shift aligns with previous research. In a 2022 study by Bluemel et al. on Atlantic wolffish populations in the North Sea off the coast of the United Kingdom, the populations were projected to have shifted northward and farther from land from 1978 to 2020 using an ensemble model. This phenomenon, driven by increased temperatures and coastal anthropogenic activity, corroborates the random forest-forecasted wolffish distributions. Conversely, in a 2016 study conducted by Bianucci et al., their model projected a retraction of suitable Atlantic wolffish habitat towards the shore off the Scotian Shelf due to decreased dissolved oxygen on the ocean floor. Despite similarities among the three models, the data used to derive them are from more consistent bottom-trawling surveys and include additional covariate information, resulting in more consistent results. In the future, it could be helpful to examine additional covariates, such as anthropogenic activity and dissolved oxygen, and to supplement this data with more consistent trawling survey data from NOAA's database.

By anticipating this northerly trend in the Atlantic wolffish population, the government can implement administrative measures to protect wolffish habitats from overfishing or destruction. Whether it's an altered

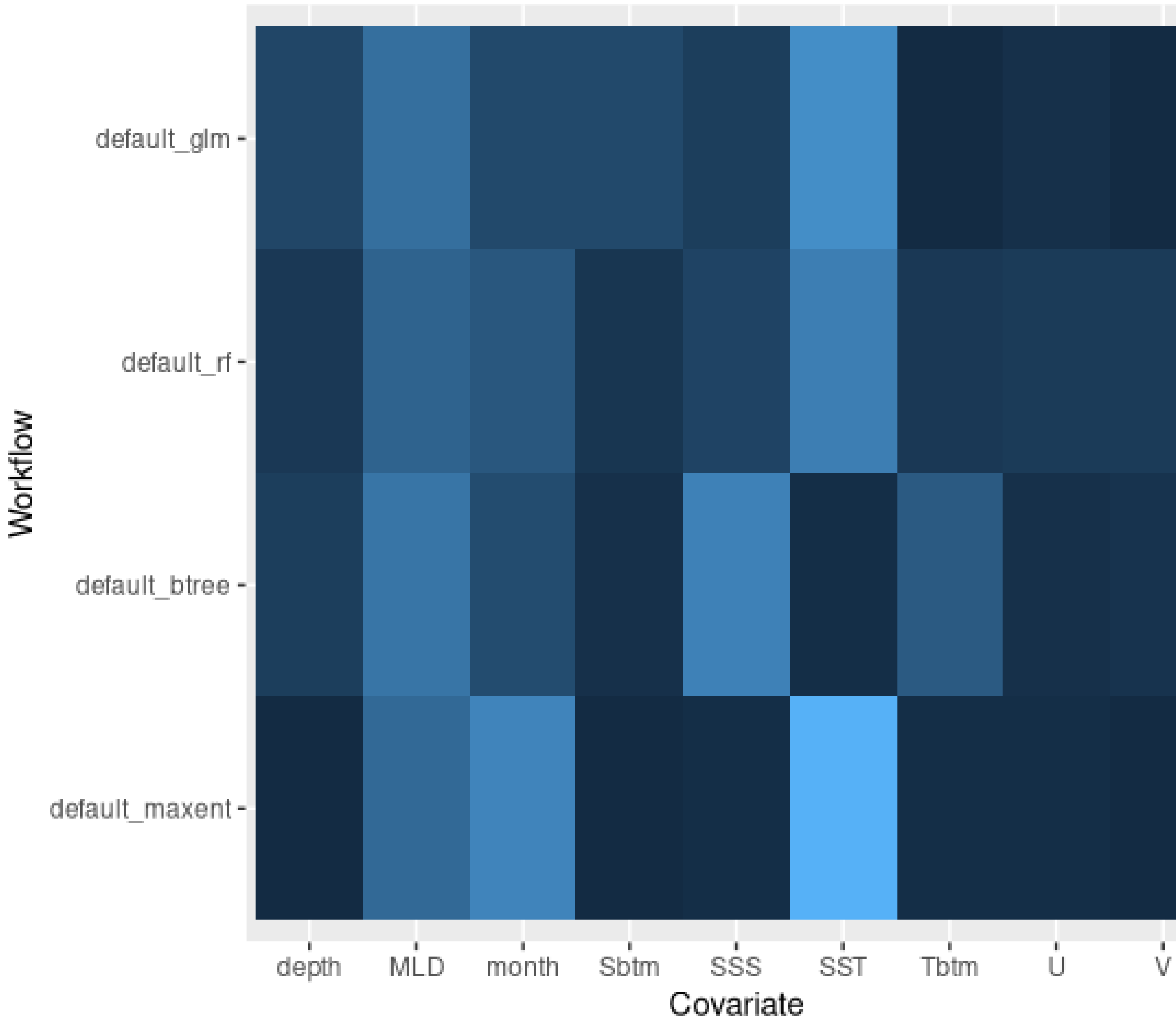


Figure 11: Figure 11) Plot of covariate contribution to model decision for four different distribution models (boosted tree, generalized linear, maximum entropy, and random forest models).

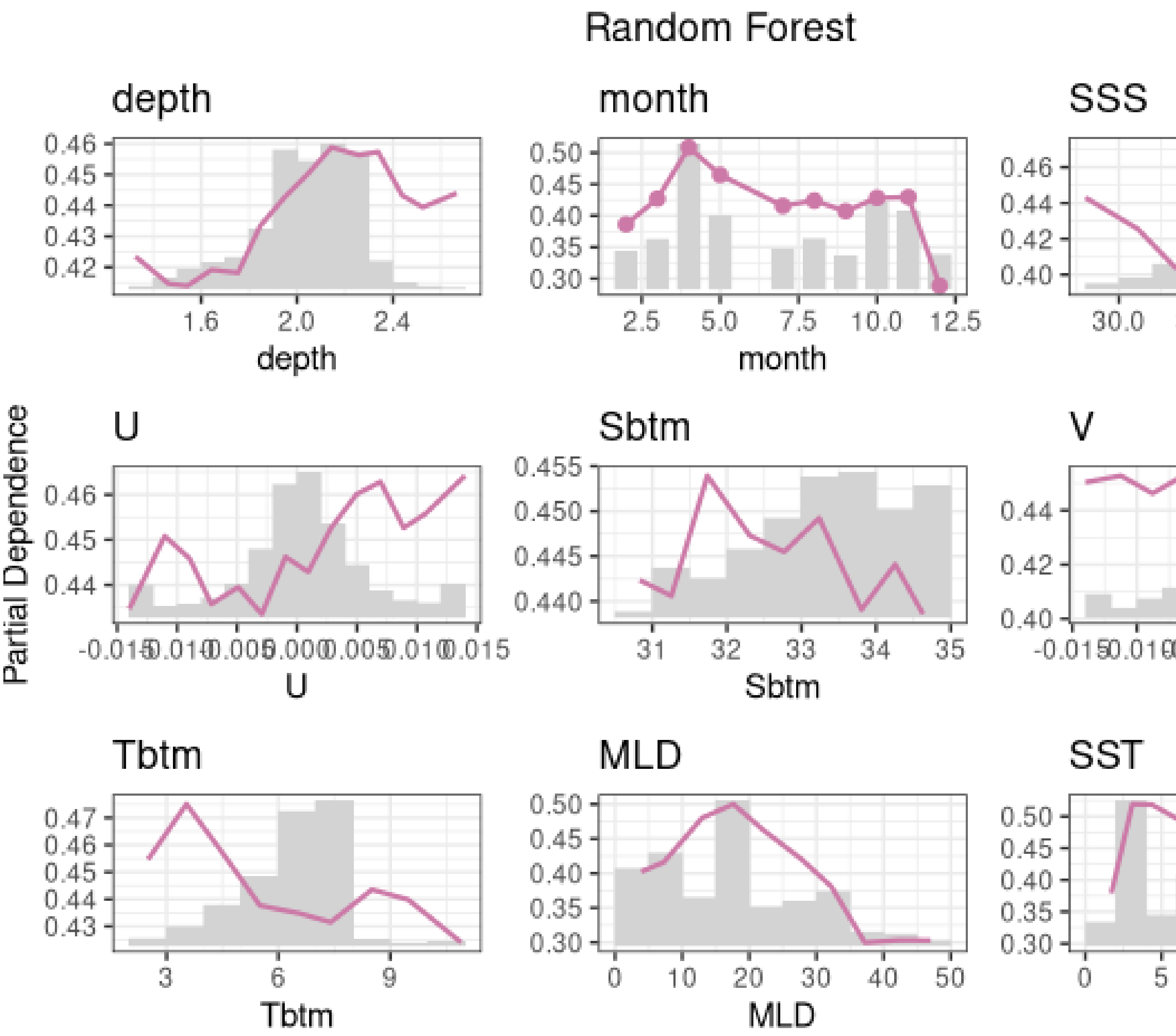


Figure 12: Figure 12) Partial dependence plot of covariate importance for the random forest model.

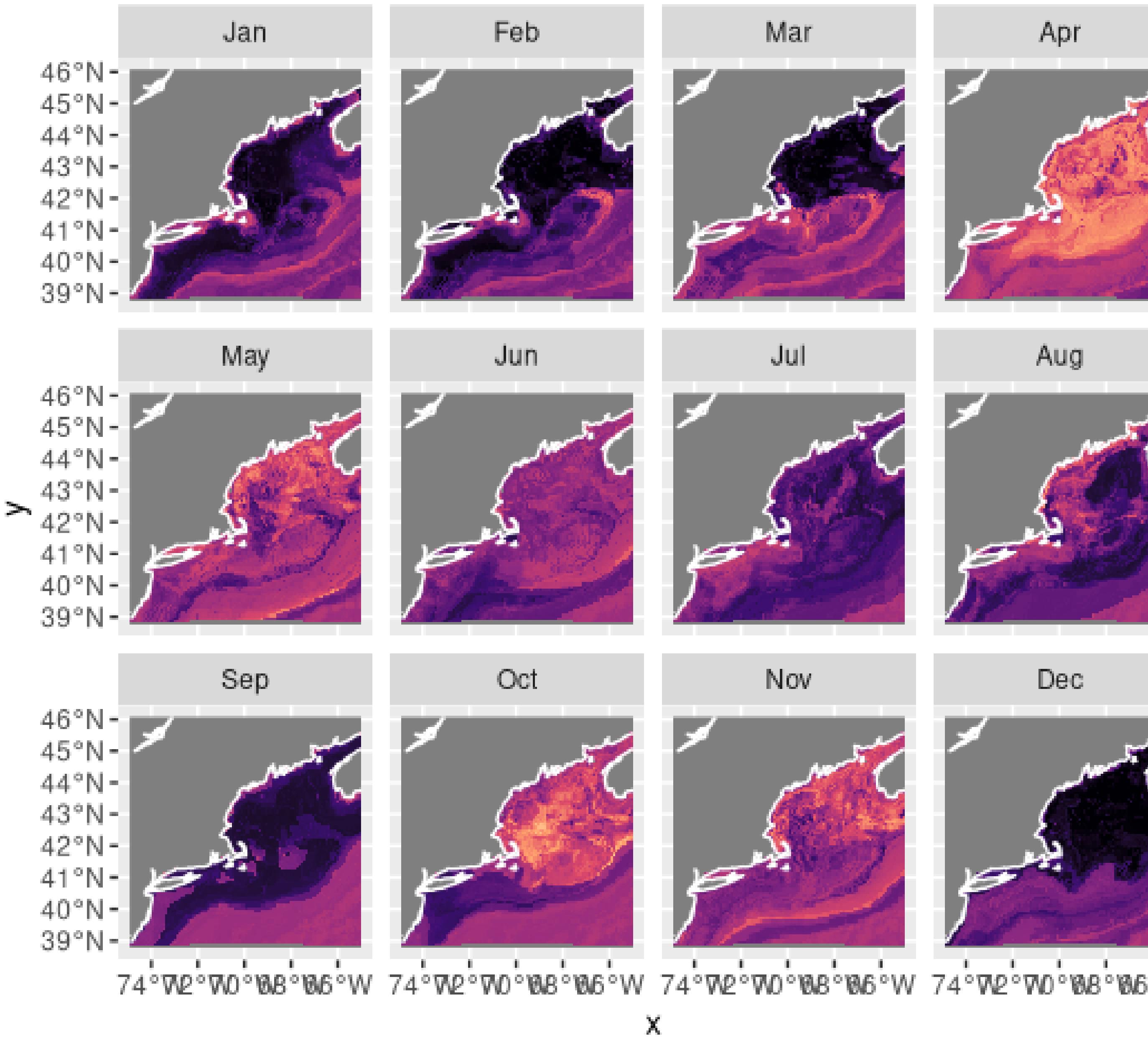


Figure 13: Figure 13) Maps of monthly spatial distribution of Atlantic wolffish using the random forest projection in the year 2025.

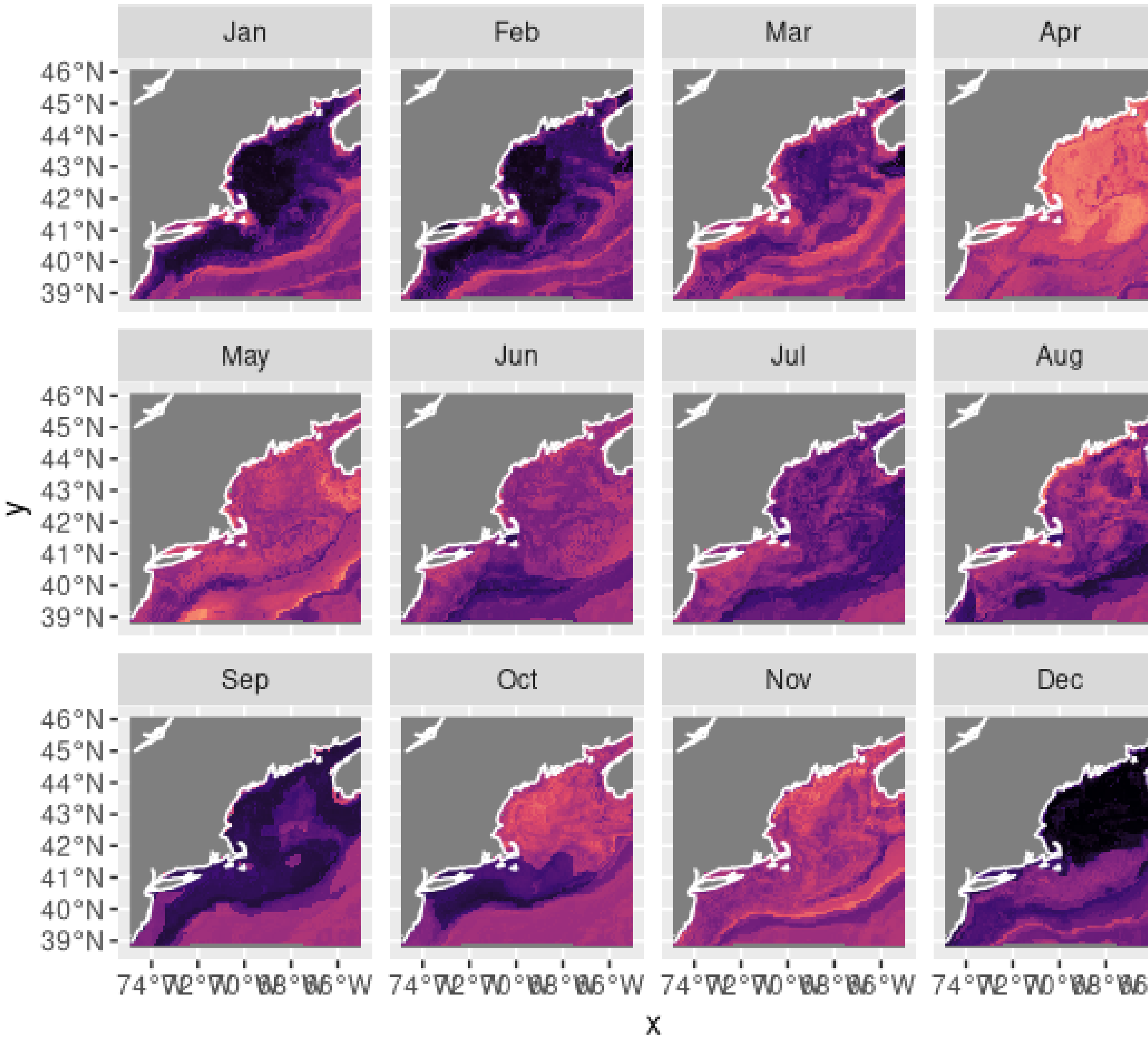


Figure 14: Figure 14) Maps of monthly spatial distribution of Atlantic wolffish using the random forest projection in the year 2055 under RCP 4.5 conditions.

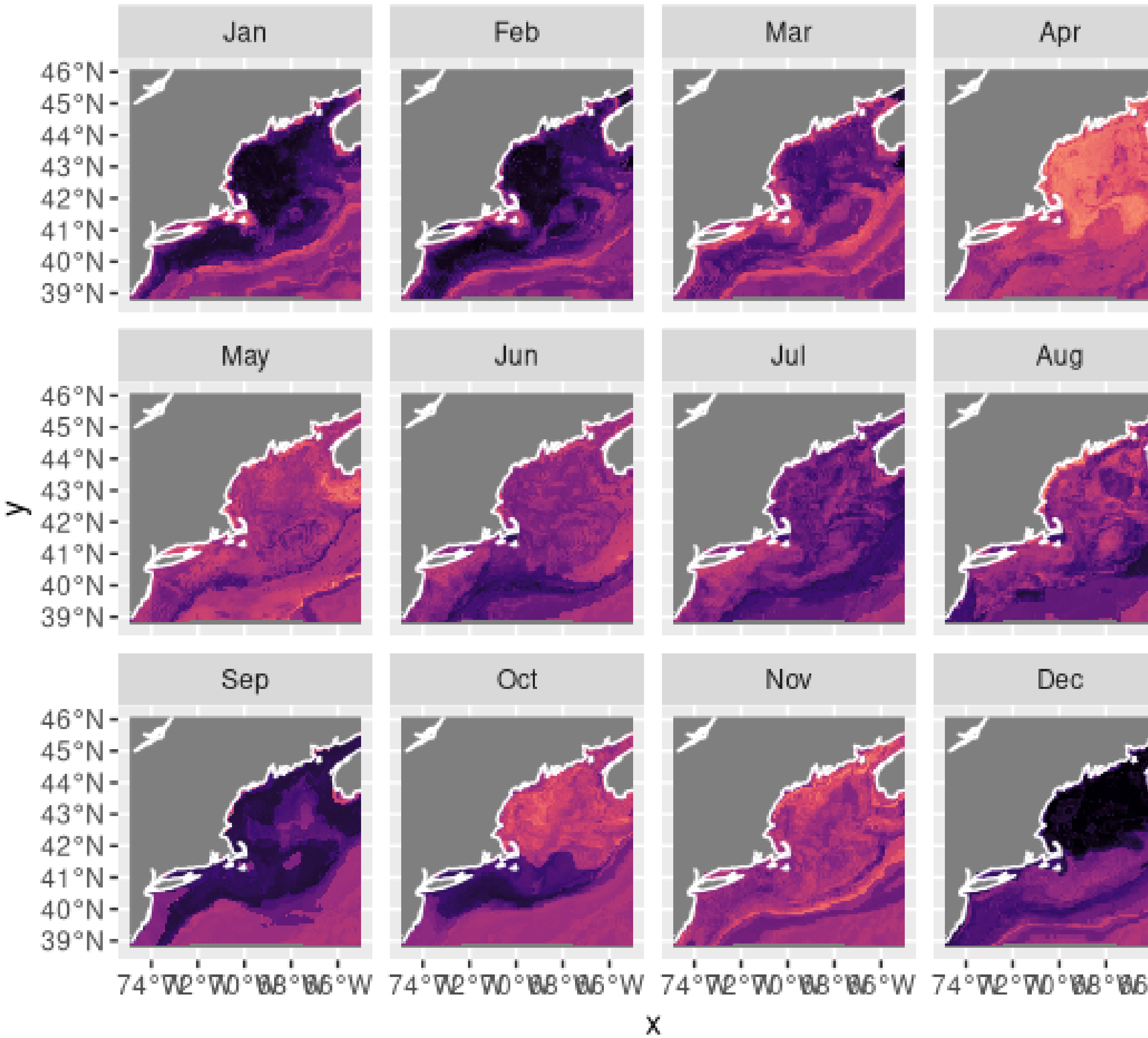


Figure 15: Figure 15) Maps of monthly spatial distribution of Atlantic wolffish using the random forest projection in the year 2075 under RCP 4.5 conditions.

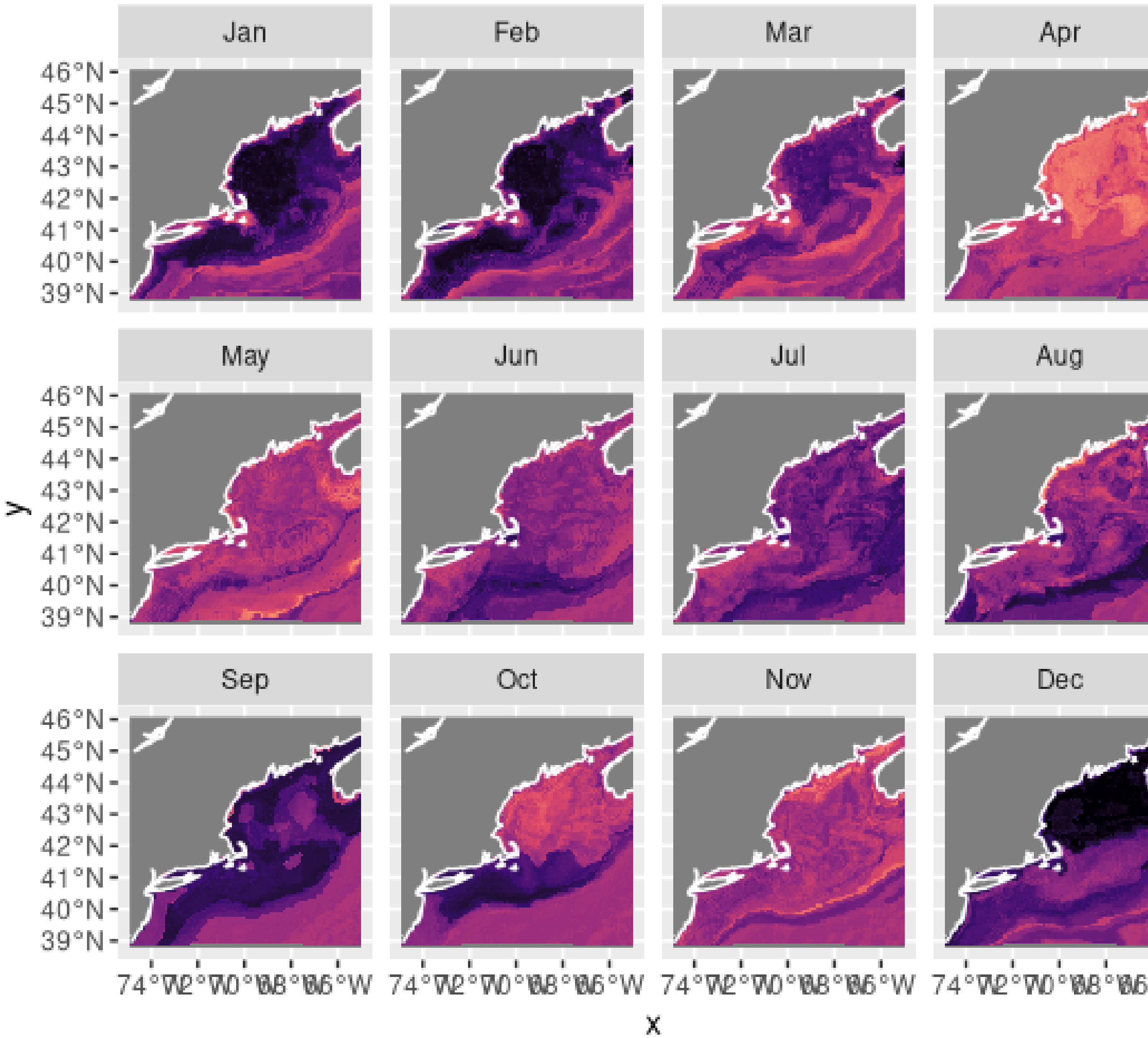


Figure 16: Figure 16) Maps of monthly spatial distribution of Atlantic wolffish using the random forest projection in the year 2055 under RCP 8.5 conditions.

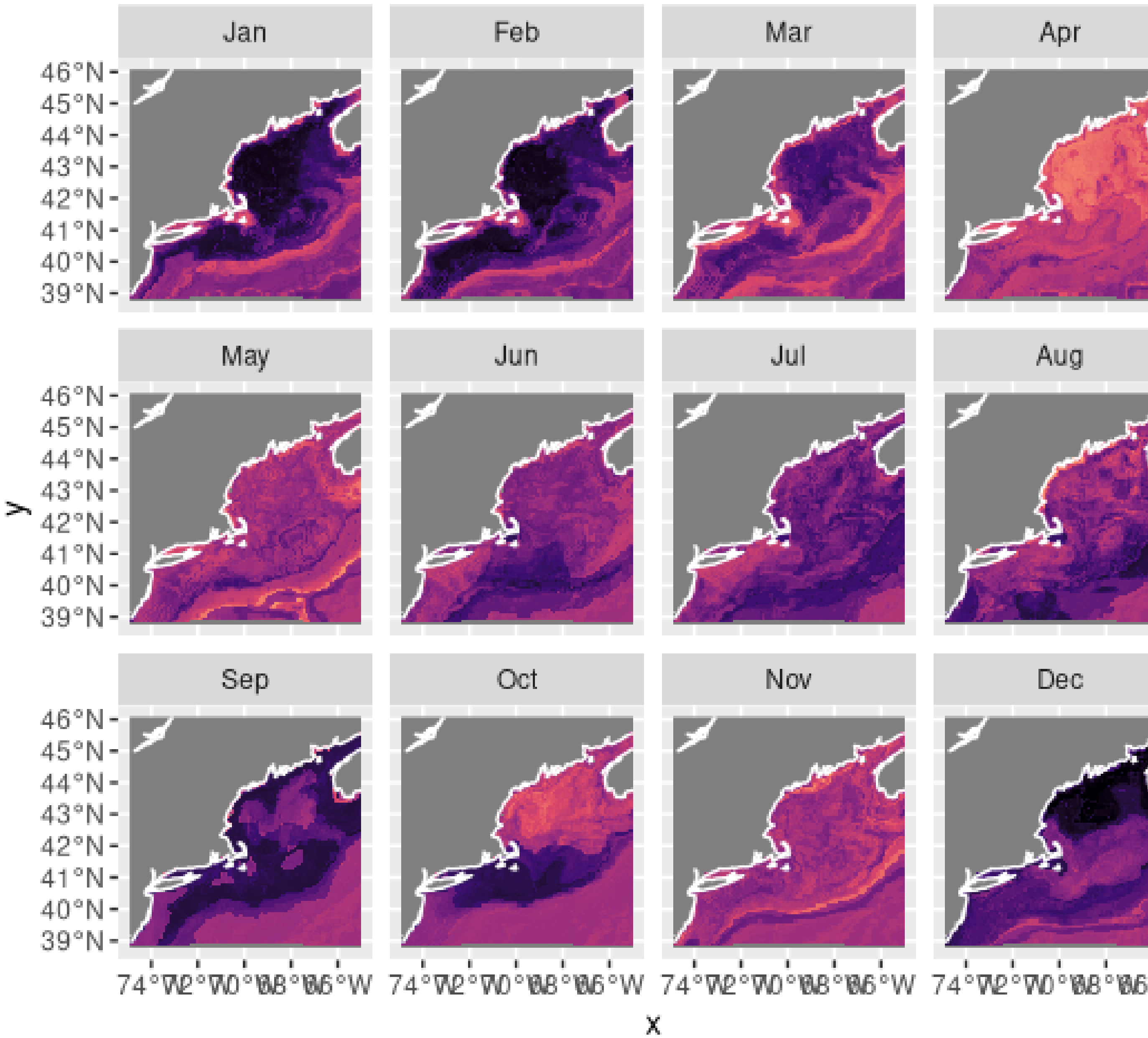


Figure 17: Figure 17) Maps of monthly spatial distribution of Atlantic wolffish using the random forest projection in the year 2075 under RCP 8.5 conditions.

lobster-trapping or scallop-dredging season, this conservation effort will help ensure the continued health of kelp forests and the species that rely on the wolffish-protected ecosystems. These northerly trends can also support the ongoing effort to establish the Atlantic wolffish as an aquaculture species. By providing insight into the wolffish's range, farmers can better understand locations suitable for sea-cage harvesting, a more sustainable way to produce food.

Despite the informative potential of the random forest podcast, the Atlantic wolffish's data deficiency limits the scope of application of these results. To combat this issue, not just with the wolffish but also with other demersal and benthic species, scientists could be employed alongside lobstermen and other types of fishermen to identify and release bycatch species. By cooperating with local fisheries, relationships can be fostered that facilitate easier communication regarding fishing guidelines and help bridge the understanding between the needs of fishermen and the scientific community.